

RISANSI INDUSTRIES LTD

Pump Testing Portal

Functional Documentation

rndtesting.risansi.com

Version: working tree as of 5 October 2026

Audience: R&D / production testing team, source teams, administrators

Status: Internal document

1. Purpose and scope

The Pump Testing Portal is Risansi Industries Ltd's internal web application for the R&D and production testing team. It replaces ad-hoc requests and Excel test sheets with one system in which source teams raise testing requisitions, the testing team runs the tests and records the results, and managers see progress, results and workload.

It does four main things:

- Intake: raise a pump testing requisition and check it against earlier tests so work is not repeated.
- Test reporting: fill in Observation Sheets and Viscosity Correction Charts with live calculations, judge them against rated Head / Capacity / Power, and print them as PDF.
- Tracking and analysis: status, deadlines, workload, pass rate, VE / ME performance against acceptance criteria, improvement projects and a testing calendar.
- Administration: users and access, bug reports, an action registry for retests and a full audit log.

The portal is available at <https://rndtesting.risansi.com>. It shares its user database with the sales portal, so one login works in both.

2. Roles and what each can do

Role	Typical person	Can do
Source Team (source)	Proposal / Research / sales engineers	Raise requisitions and see only their own. Edit their own requisition's intake fields (model, category, EC/quotation no., offer date, responsible person, source team, dates, rated values). Cannot touch status or testing results. Cannot submit, edit or delete reports.
Testing Team (testing)	R&D / production testing staff	Raise requisitions. Start Testing, mark Retest Needed and submit test reports. The only role that can edit a report (within 10 days) or delete one, and the only role that can reassign the Responsible Person from the Testing Summary list. Sees every requisition.
Central Admin (central-admin)	Senior reviewer	Raise requisitions. Start Testing and submit test reports (as admins can). Assign Retest on reports that missed a rated target and open the Action Registry. No access to user management or access requests. Cannot edit or delete reports.
System Admin (admin)	Portal administrator	Raise and process requisitions, submit reports, Assign Retest and the Action Registry, plus the admin-only features: manage users (roles, passwords, activate, deactivate, delete), approve or reject sign-up requests, delete requisitions, create and edit calendar events, view bug reports, the Audit Log and this documentation. An admin cannot edit or delete a test report; that is reserved for the Testing Team.

Everyone who is signed in can view the Testing Calendar. Only a System Admin can create, edit or delete calendar events.

3. Signing in and account lifecycle

3.1 Requesting access

A new user opens Request Access, enters name, email and picks Source Team or Testing Team. The account is created as Pending. Central Admin and System Admin accounts are never self-service; an administrator sets them.

3.2 Approval

A System Admin sees pending requests at the top of Users & Access and approves or rejects each one. Approving makes the account Active. Admins can also add a user directly.

3.3 First sign-in and passwords

- Every account starts with a forced password change. A non-dismissible dialog appears after sign-in until a new password is saved.
- A session lasts 12 hours, then the user signs in again.
- Admins can reset any user's password. Users can change their own from the profile menu.
- Deactivated or rejected accounts cannot sign in, and an existing session stops working on its next request.

4. Screens and features

4.1 Dashboard (Overview)

The landing page. A greeting, a scope switch (All requisitions / Created by me), a date filter (Today, This week, This month, 7 / 30 / 90 days, All time or custom dates), and three extra filters: Category, Model and Status. Every card and chart below respects all of them.

- KPI cards: Totals (requisitions + reports, pendency), Pending (requisitions and overdue), In progress (In Testing, Retest Needed), Completed (closed requisitions + reports, pass rate), Open & turnaround.
- Monthly or daily activity chart: requisitions raised, reports filed, requisitions closed.
- Requirement results: how many reports met or missed their rated target, and which parameter (Head, Capacity, Power) they miss.
- Workload per responsible person, requisitions by raiser type, by source team and by category, a category-by-month heat map, recent reports and the nearest upcoming deadlines.
- Click any figure, bar or cell: a pop-up list opens right on the dashboard showing exactly those requisitions or reports, with search, "Load more", links to each record and an "Open full page" button. Ctrl-click opens the full page in a new tab.
- Against Improvement Project: choosing this category in the dashboard filter (or clicking a link for it) shows the list of improvement models from VE & ME Performance instead of the figures: each model with its reports below acceptance and its latest VE / ME. It ignores the date, scope and status filters; the Model filter narrows it.

4.2 New Requisition

A form to raise a testing request. Key fields: Model, Category, EC / Quotation / Offer No. (label and format hint change with the category), offer date, Responsible Person, Source Team, date of requisition, quantity, QTH, power (HP / kW), head, RPM, motor RPM (1440 or 960), required capacity, remarks and attachments.

- Categories: Against Pump Testing Project, New Die Pin, Die Pin Rework, Quotation Test, R&D Trials, EC Based and Improvement Project.
- Automatic assignee: the Responsible Person is set from the category. EC Based requisitions go to Vikash; every other category goes to Sachin. The Testing Team can reassign afterwards.
- Earlier-test check: for a model, the portal can list the test reports already on file for that pump (matched on the model name, ignoring capitals) and tells you whether it has been tested before, so a result can be

reused instead of re-testing.

- Requisitions get a readable number (REQ-000123). The target date defaults to date of requisition + 7 days unless set explicitly.
- Attachments can be added to a requisition and downloaded later.

4.3 Testing Summary

The working list of all requisitions, with status tabs (All, Pending, In Testing, Retest Needed, Closed) and a filter bar: search by EC / quotation number, Model, Deadline (open, overdue, due in 5 days), Category, Raised by, Source team, Responsible, Submitted by, Retest, Month and a date range with shortcuts (Today, Yesterday, Last 7 days, Last 30 days, This month, This year). Filters that are active are highlighted and one button clears them all.

- Rows load automatically as you scroll (25 at a time).
- Testing Team users can reassign the Responsible Person straight from the row.
- Closed requisitions show View Report and View Curve; reports that missed a rated target are flagged with a warning.
- On Closed or category views a Met / Not Met filter appears for the linked reports.
- Admins see a Delete button; deletion is permanent and recorded in the audit log.

4.4 Requisition detail and edit

Shows every field, attachments and linked reports. The workflow is:

- Pending: if the pump has no earlier reports, the tester presses Start Testing and the status becomes In Testing. If earlier reports exist they are listed and the tester chooses Retest Needed: Yes (status Retest Needed) or No (status In Testing).
- In Testing / Retest Needed: when physical testing is done the tester presses Fill Test Report. Submitting the report closes the requisition automatically.
- Closed: the page shows the report and links to it.
- Anyone except a Source Team user can run these steps. Source users can edit only the intake fields, and only on their own requisitions.

4.5 Test reports

Two report formats share the same structure and are field-for-field identical:

- Observation Sheet: capacity measured by V-notch, Barrel or Flow Meter.
- Viscosity Correction Chart: the same measurement methods plus liquid-viscosity-corrected efficiency figures.

Header: gearbox and motor information, PO / EC number, NPSHa, rated capacity, head and RPM, specific gravity, viscosity (CPS), K for the given CPS, reference voltage and current, V-notch baseline, vibration test and run summary, Witness / Inspector / Recorder and remarks. Below it is the table of test points.

- Live calculation: slip, capacity, power and efficiencies are computed on screen using the formulas from the original Excel workbook ("NEW PUMP TESTING FORMATE FOR DIGITAL"). Total Run (stop - start time) and Total Rise (maximum bearing temperature - ambient temperature) are calculated, not typed.
- Auto-fill: a Viscosity Chart linked to a requisition, or whose model matches an existing Observation Sheet, copies the shared fields and test points from that sheet into any still-empty fields. Nothing the tester already typed is overwritten.
- Requirement check: a report is Met when the highest Head and highest Capacity reach the rated values and the highest Power stays at or below the rated power. A missed target flags the report; it does not block saving.

- Who can do what: anyone except a Source Team user can submit a report. Only the Testing Team can edit or delete one, and editing is allowed for 10 days after submission.
- Report detail page lists test points sorted by head, lowest first, and shows the performance curve.
- Export PDF: the Export PDF button opens the browser's print view with the Risansi header; Witness / Inspector / Recorder print as a Role / Name / Signature table with a blank signature line.

4.6 Report Archive and Report Compilation

- Report Archive: all reports grouped by physical pump, with a search box, date range, category and Met / Not Met filters. Shows each report's maximum VE and ME. About 250 historical reports were imported from old Excel / PDF sheets; they are matched by the shop nickname of the pump (for example H-85).
- Report Compilation: every pump model with its reports and requisitions side by side; open a model for its complete dashboard.
- A report's category is read from the report itself (the category note in its remarks, or its EC / quotation number) or from its linked requisition.

4.7 VE & ME Performance

For every pump model (including those never tested) the page shows the best and latest Volumetric (VE) and Mechanical (ME) efficiency against the acceptance table supplied by the testing team, how they have changed since the first test, and every test in date order.

- Acceptance table: H, 2H, L and L6 series, each with a required VE % and ME %.
- Improvement Projects: any report in the Improvement Project category, and automatically any report whose VE or ME is below its model's acceptance value, is listed here, grouped by model. Each model is a drop-down. Open it to read its reports from oldest to newest, so you can follow the model up or down: every row shows the change in VE and ME since the previous test (up arrow better, down arrow worse), a verdict (Improved, Declined, Mixed, No change or First test) and which test it was compared with. A summary at the top gives the required VE and ME, the first test, the latest test and the overall change. Click a report number to open the report.
- A model is left out of this list as soon as ANY ONE of its reports meets BOTH its VE and its ME acceptance value, however many of its other reports fall short, and all of its reports leave with it. A note under the list names the models that are not listed. Every test of those models stays in the By model table, still shown in red where it was below acceptance.
- Impossible readings (VE above 150 % or ME above 100 %) are flagged for correction and never counted as a best, latest or accepted result.

4.8 Assign Retest and Action Registry

On a report that missed its rated Head, Capacity or Power, Admin and Central Admin see Assign Retest (on the report page and in Report Compilation). The dialog lists what was missed (rated versus measured) and lets them write action points. Submitting does two things: raises a fresh Pending requisition with Retest Needed already set, so it drops back into the testing queue, and records an entry in the Action Registry. The Action Registry lists every such entry with who assigned it and who originally raised the test.

4.9 Testing Calendar

A weekly grid with one row per responsible person (Sachin, Vikash) and a column per day, or a month view. Cards show the title, EC / quotation number or model, time and a status of Planned or Completed. A header shows the number of events and the percentage completed, and a filter selects Planned or Completed. Everyone signed in can view it; a System Admin creates, edits and deletes events. The event form has a searchable EC / Quotation No. field and requires an assignee.

4.10 Notifications and the profile menu

- A bell lists requisitions assigned to you that are due within five days or overdue.
- Admins see an unread count for bug reports and a count of pending access requests.
- Light and dark themes are supported. Change Password and Sign out are in the profile menu.

5. Administration

5.1 Users & Access

A list of all users with search and status / role filters: name and email, role, status, whether active, requested and reviewed dates, and actions to edit, deactivate or activate, reset password and delete. Deleting a user keeps the history readable; requisitions keep the submitter's name.

5.2 Bug Tracker

Any user can file a bug or feature request from the Report a Bug button (title, description, severity, page, optional screenshot). Admins see them as cards in Open, In Progress and Resolved columns, with totals above, filters for type, severity and search, and drag-and-drop to change status. Opening a card marks it read.

5.3 Audit Log

An admin-only record of who signed in and what they did. Header figures show the last 24 hours (logins, failed sign-ins, active users, actions). Tabs: Overview, Usage by User, Activity, Logins & Sessions and Access Changes. The Overview shows adoption (active users, never signed in, dormant, sessions, active hours, records touched, sign-ins), an activity trend, event types, when people work (weekday-by-hour heat map), action breakdown, per-user daily activity grids, devices, browsers and IP addresses. A PDF report can be generated for the selected range.

How Active Time is calculated: every recorded action by a user (signing in, saving, creating, updating or deleting) has a time; failed sign-ins are ignored. A user's action times are put in order and the gap between each one and the next is measured. Gaps of 15 minutes or less are added up as active time; a longer gap means the user stepped away and counts as idle. Example: actions at 10:00, 10:04, 10:09 and 11:30 give 4 + 5 = 9 minutes. The 81-minute gap is idle.

5.4 Documentation

An admin-only page in the Admin section of the sidebar. It offers this functional documentation and the technical documentation as PDFs to open or download. The files are stored privately on the server and are only sent to a signed-in admin.

6. Business rules at a glance

Rule	Definition
Requirement met	Highest recorded Head $\geq$ rated Head and highest Capacity $\geq$ rated Capacity, and highest Power $\leq$ rated Power. A report with no rated target is neither met nor missed.
Pass rate	Reports that met their target / reports that have a target to judge.
Pendency	Requisitions with status Pending.
Completed	Closed requisitions plus every filed report (a filed report is finished testing).

Open	Pending, In Testing and Retest Needed.
Overdue	Open requisition whose target date is earlier than today. Target date defaults to date of requisition + 7 days.
Report edit window	10 days from submission, Testing Team only.
Default assignee	EC Based requisitions go to Vikash, all other categories to Sachin.
VE / ME acceptance	Best VE $\geq$ model's VE % and best ME $\geq$ model's ME %. Models not in the table are not judged.
Improvement Project	Category is Improvement Project, or VE / ME below acceptance, unless at least one report of that model meets both the VE and the ME acceptance value (then the model is not listed at all).
Session length	12 hours.
List size	25 rows per page (summary lists load as you scroll); 50 groups per page in the archive and compilation.
Test points order	Head ascending, RPM as tie-break.

7. Reports and exports

- Test report PDF: print view of the Observation Sheet or Viscosity Correction Chart (use the browser's Save as PDF).
- Audit report PDF: generated by the server for the selected date range.
- Charts and tables on the dashboard and audit pages have a Table view for reading every value.

8. Glossary

Term	Meaning
EC No.	The reference number entered on a requisition. For EC Based requisitions the format is EC/YEAR/1/So No./EC No.
Quotation / Offer No.	Format RIL/QT/TSM CODING/CATEGORY OF ITEM/OFFER.
VE	Volumetric efficiency, a whole percentage.
ME	Mechanical efficiency, a whole percentage.
K for given CPS	Viscosity correction factor used in the slip and efficiency calculations.
V-notch / Barrel / Flow Meter	The three ways capacity is measured during a test.
Retest Needed	Status for a requisition whose test must be repeated.
Action Registry	Permanent list of retests assigned, with the action points.

9. Known limitations

- Report edits are limited to 10 days after submission.
- Responsible Person is limited to the two named testing engineers.
- Active time depends on what the portal records: this portal logs sign-ins and creates / updates / deletes, not every click, so active time under the 15-minute rule is a conservative figure.
- The portal has no automated test suite; changes are checked by type-check and manual testing.